Mock Interview Edition

Search Ranking: From Boosted Trees to a Neural Ranker

18 scenarios with their follow-up probes and no answers. Use it to rehearse aloud, with a partner, or against a timer, then check yourself against the Study Edition.

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HOW THIS EDITION IS ORGANIZED

Section A, round plans: 5 ready-made sessions of five scenarios each, by theme. **Section B, question sheets:** every scenario followed by all of its probes, in order, each with its own notes box. The Study Edition page that holds the answers is printed under each scenario. **Section C, score sheet:** one row per scenario to track attempts. Question numbers and part numbers are identical in the Study Edition.

THREE WAYS TO USE IT

1. With a partner. The partner reads the scenario aloud, then asks the probes one at a time, moving on only after you have answered. **2. Alone, out loud.** Cover everything below the current probe with a sheet of paper, answer aloud for about two minutes, then slide the paper down to the next probe. **3. Alone, in writing.** Jot your key points in the notes box under each probe before looking at the next one, then compare with the Study Edition page printed under the scenario.

HOW TO GRADE A PROBE

Solid: you stated the main point, gave a number or a concrete example where one was possible, and said how you would check it. **Partial:** right direction but vague, or a number without the reasoning. **Missed:** you did not reach it, or reached the wrong conclusion. The last probe of every scenario asks for the general lesson in one sentence. Suggested pacing is about 12 minutes per scenario. Reading the answer only after you have committed to your own is what makes the practice work.

A NOTE ON THE PACING, GRADING AND ROUNDS

The 12-minute pacing, the three-level grading and the round compositions are the author's suggestions for practice, not an industry standard. The scenarios are written for this document and are not taken from a real company question bank. Where a scenario quotes a figure, the Study Edition says where it comes from.

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Every scenario, by part

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SECTION A

Round plans

Each round has five scenarios at about 12 minutes each, a 60-minute session. Mix and match freely. Click any scenario to jump to its question sheet; the number on the right is the page.

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SECTION B

Question sheets

Each scenario is followed by its probes in order. Indentation shows how deep the follow-up goes.

PART 1

Framing the problem: labels, the metric, the plateau and the sizing

Choosing a label and a metric for a marketplace, deciding what to do when a good model stops improving, and finding which resource binds.

01 What is the label, and who produces it?

FRAMING

SCENARIO

The paper trains a ranker for listing search. A colleague proposes using clicks on listing pages as the label because they are plentiful. The paper uses bookings. Explain the choice, and say exactly where the training examples come from.

OPENING PROBE

What does the paper train on?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Why a booking and not a click?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Who is missing from the data?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What else shapes the data?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 6

02 The model has stopped improving. What now?

PLATEAU

SCENARIO

A tree-based ranker replaced a manual scoring function and gave one of the largest step improvements in bookings in the company's history. Years later, A/B tests keep coming back neutral. A director asks you to move to a neural network. Plan the move.

OPENING PROBE

What does the paper say about the plateau?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What went wrong at first?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What was the first stage for?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What was the schedule?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 7

03 Compute NDCG by hand

CALCULATION

SCENARIO

An interviewer says: a booked listing moves from position 2 to position 1 in one search, and another booked listing moves from position 10 to position 9 in another. Which improvement matters more to NDCG, and how much more? Use the standard discount of 1 divided by log2 of (position plus 1).

OPENING PROBE

What are the discounts?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does each move gain?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does that imply for training?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What can't you read from the paper?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 8

04 Where does the cost sit?

ESTIMATION

SCENARIO

The paper's final model takes 195 input features, then 127 and 83 ReLU units, trained on 1.7 billion pairs. It gives no search rate or inventory size. A manager asks whether the cost is in scoring, in training compute, or somewhere else. Use 2,000 candidates a query and a 4 KB static feature record per listing (both illustrative).

OPENING PROBE

What does scoring cost?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does a training pass cost in compute?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does it cost in bytes?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What happens at ten times the data?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 9

PART 2

Features and data: normalization, smoothness and the ID that failed

Why the same features that suit a tree model break a neural network, what a distribution plot reveals, and why a per-item embedding needs data per item.

05 Why did the same features break the network?

NORMALIZATION

SCENARIO

The first neural attempt used exactly the features the tree model used. The paper says it went very badly: the loss saturated mid-training and more steps did nothing. Explain the cause and the fix, and show the transform on prices.

OPENING PROBE

What does a tree use from a feature?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does a network do with it?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What is the fix?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Apply it to prices.

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 10

06 Plot the distribution first

DATA QUALITY

SCENARIO

The paper says it cares about smooth feature distributions, and gives three reasons. A teammate asks why a distribution plot is worth an engineer's time when range checks exist. Answer with the paper's examples.

OPENING PROBE

What bugs did the plots catch?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does smoothness do for generalization?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What did a lumpy feature reveal?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How was location handled?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 11

07 Embed every listing ID?

FEATURES

SCENARIO

Neural networks make it easy to give every listing its own learned embedding, as recommenders do for videos. The paper tried it. Predict what happened, explain why, and give the arithmetic.

OPENING PROBE

What happened?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Why does it work for videos and words?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What limits a listing?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What did the team do instead for location?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 12

08 Review the price transform

CODE REVIEW

SCENARIO

A colleague writes this price feature for the neural ranker, following the paper's power-law transform. Offline NDCG is flat and the model ignores price. Premium and budget markets look alike to the model. Find the faults.

```
import math

def normalize_price(prices):
    # prices: nightly prices of the candidate listings for ONE query
    median = sorted(prices)[len(prices) // 2]
    return [math.log((1 + p) / (1 + median)) for p in prices]
```

OPENING PROBE

What is the median taken over?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Can serving reproduce it?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What else is missing?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

State the fix.

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 13

PART 3

The models: trees, LambdaRank, the hybrid and the deep network

When to leave a tree model, what a pairwise weighted loss does, what the hybrid taught, and how to read a learning curve.

09 Trees or a neural network?

DECISION

SCENARIO

Your team has a tree ranker that works and has plateaued. The paper ends with a wholehearted yes to deep learning and also says trees are arguably at par for smaller problems. A colleague says that means the paper contradicts itself. Resolve it, and give a rule for your own team.

OPENING PROBE

What is the paper's recommendation?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What did the neural network add here?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What did it cost?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What is a rule for your team?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 15

10 What does LambdaRank do to the loss?

ALGORITHM

SCENARIO

The paper's breakthrough came from combining a neural network with the idea behind LambdaRank. A candidate says: that proves trees cannot be trained for NDCG. Explain what the loss does, compute an example, and challenge the claim.

OPENING PROBE

What are the two changes?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Compute one pair.

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

Where do the ranks come from?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Does the paper show trees cannot do this?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 16

11 Combine three models, then drop two

ARCHITECTURE

SCENARIO

The team's other models, a boosted tree model and a factorization machine, matched the neural network on test data and still ranked different listings highly. They built a network that takes the tree leaves and the FM's prediction as inputs. Explain the reasoning, the cost, and what happened next.

OPENING PROBE

Why combine them?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How were they combined?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What did it cost?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What replaced it?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 17

12 Review the LambdaRank loss

CODE REVIEW

SCENARIO

A colleague writes this pairwise loss for one search, with the booked listing at index 0, and says it implements the paper. The loss is sometimes infinite or not a number, and the model does not improve at the top of the list. Find the faults.

```
import numpy as np

def lambdarank_loss(logits):
    # logits: (n,) scores for one search; logits[0] is the booked listing
    order = np.argsort(-logits)                  # listing indices, best first
    discount = 1.0 / np.log2(order + 1)
    weights = np.abs(discount[0] - discount[1:])
    diff = logits[0] - logits[1:]
    return np.mean(weights * np.log1p(np.exp(-diff)))
```

OPENING PROBE

What does argsort give here?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What goes wrong at the top?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What about gradients?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

State the fix, and a test.

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 18

13 Read a learning curve

EVALUATION

SCENARIO

The paper plots NDCG on training and test data against the number of training pairs, and says that training on 1.7 billion pairs closed the generalization gap. A manager asks what that means and whether to buy more data or a bigger model. Explain.

OPENING PROBE

What does a learning curve show?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 5

What did the listing ID experiment look like?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 5

What does this say about data and model size?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 5

What did more data cost?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 5

What does the paper leave open?

YOUR NOTES: key points you would make

FOLLOW-UP 5 OF 5

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 19

PART 4

Serving, evaluation and operations

Feeding a GPU, what a static embedding is for, scoring in another language, trusting offline numbers, and understanding what the model uses.

14 Feed the GPU

TRAINING PIPELINE

SCENARIO

Training reads CSV files and copies the values through feed_dict. GPU utilization is near 25%. The paper retooled the pipeline with Protobufs and the Dataset API and reports a 17 times speedup with utilization near 90%. A manager asks whether raising utilization from 25% to 90% explains the 17 times.

OPENING PROBE

What is the ratio?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

So what else changed?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What did it enable?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

How would you find the bottleneck yourself?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 21

15 A static embedding: for training, not serving

PIPELINE

SCENARIO

Many listing features rarely change: location, bedrooms, amenities and guest rules. The paper packs them into a non-trainable embedding indexed by listing ID and keeps it in GPU memory. A colleague draws it as a serving cache with write-through updates. Explain what it is and what it costs.

OPENING PROBE

Where does the paper put it, and why?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

How big is it?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does it cost?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

Is it a serving cache?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 22

16 Score a TensorFlow model inside a Java server

SERVING

SCENARIO

Search runs in a Java server that does retrieval and scoring, and the models are trained in TensorFlow. The paper found no efficient way to score the models in Java and wrote its own library. A reviewer asks whether that was wise. Weigh it.

OPENING PROBE

What does a call across languages move?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What does the paper say?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What does the custom library take on?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does the paper not say?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 23

17 Trust the offline number?

EVALUATION

SCENARIO

A change to the regularization raises offline NDCG by about 1%. Your team wants to ship it. The paper has this exact case. Say what to do, and what else the paper tried.

OPENING PROBE

What happened with the 1%?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

What else behaved that way?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What did the paper do about dropout?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What should the team do?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 24

18 Which features matter, and what is not covered?

ANALYSIS

SCENARIO

The paper says estimating feature importance was where the team took a step back with neural networks. It tried four methods. Explain why three of them misled, what the fourth showed, and what the paper leaves out.

OPENING PROBE

Why did score decomposition and ablation mislead?

YOUR NOTES: key points you would make

FOLLOW-UP 1 OF 4

Why did permutation mislead?

YOUR NOTES: key points you would make

FOLLOW-UP 2 OF 4

What did TopBot show?

YOUR NOTES: key points you would make

FOLLOW-UP 3 OF 4

What does the paper not discuss?

YOUR NOTES: key points you would make

FOLLOW-UP 4 OF 4

Reframe it: what is the general lesson, in one sentence?

YOUR NOTES: key points you would make

Answers: Study Edition, page 25

SECTION C

Score sheet

Write how many probes were solid out of the total, per attempt. Revisit anything under half after reading the Study Edition's concepts box.

No.	Scenario	Probes	Attempt 1	Attempt 2	Attempt 3
01	What is the label, and who produces it?	5			
02	The model has stopped improving. What now?	5			
03	Compute NDCG by hand	5			
04	Where does the cost sit?	5			
05	Why did the same features break the network?	5			
06	Plot the distribution first	5			
07	Embed every listing ID?	5			
08	Review the price transform	5			
09	Trees or a neural network?	5			
10	What does LambdaRank do to the loss?	5			
11	Combine three models, then drop two	5			
12	Review the LambdaRank loss	5			
13	Read a learning curve	6			
14	Feed the GPU	5			
15	A static embedding: for training, not serving	5			
16	Score a TensorFlow model inside a Java server	5			
17	Trust the offline number?	5			
18	Which features matter, and what is not covered?	5			